SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Vedala Teeja (192526013 3) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	1	2	2
Concept Mapping Structure	25%	2	2	2
Linking&Relationships	20%	1	1	1
Application&Analysis	20%	2	2	2
Explanation&Clarity	15%	1	0	1

Total Marks 23 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → GlobalDelivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → ErrorCorrection

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔ → DataLink ✔ → Network ✘ → Transport ✘ → Application ✘

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Linking&Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application& Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly

Name:- Veetala Jeevan

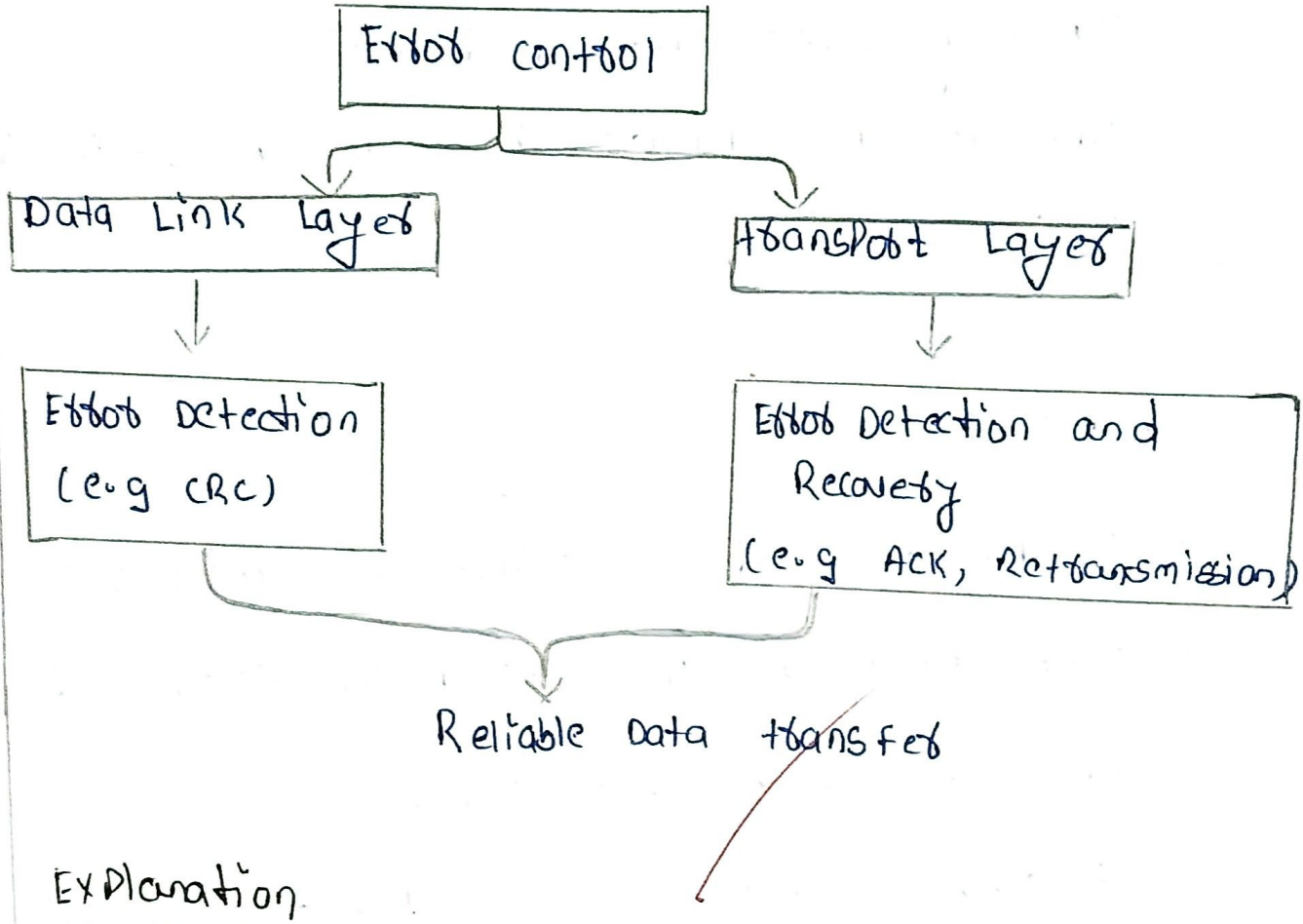
Reg no:- 192526013 3

CSA0709

CO1, AT2

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Error control in OSI Layers



Explanation

1) Data Link Layer - detects errors in frames using techniques like CRC.

2) Transport Layer - uses acknowledgements and retransmissions to recover of corrupted data

3) Improves reliability in data transmission

Transport Layer

1) The transport layer supports communication applications on different devices

2) TCP provides reliable data delivery using acknowledgements and retransmissions

3) UDP does not provide the same built-in reliability as TCP

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2) Explanation

1) It provides network services to applications

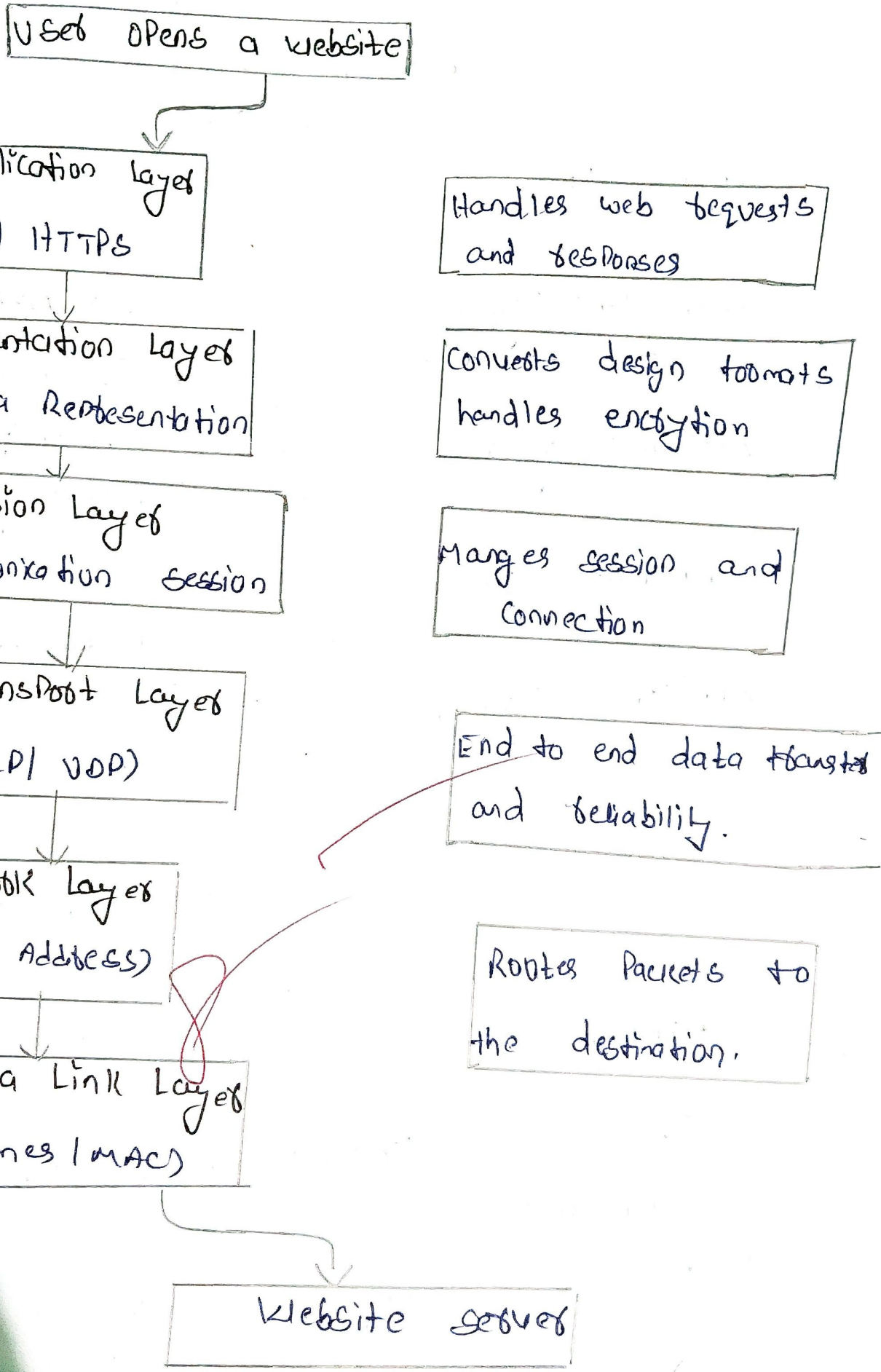
2) web browsers use HTTP or HTTPS to communicate with web servers

3) It sends request for web pages and receives responses.

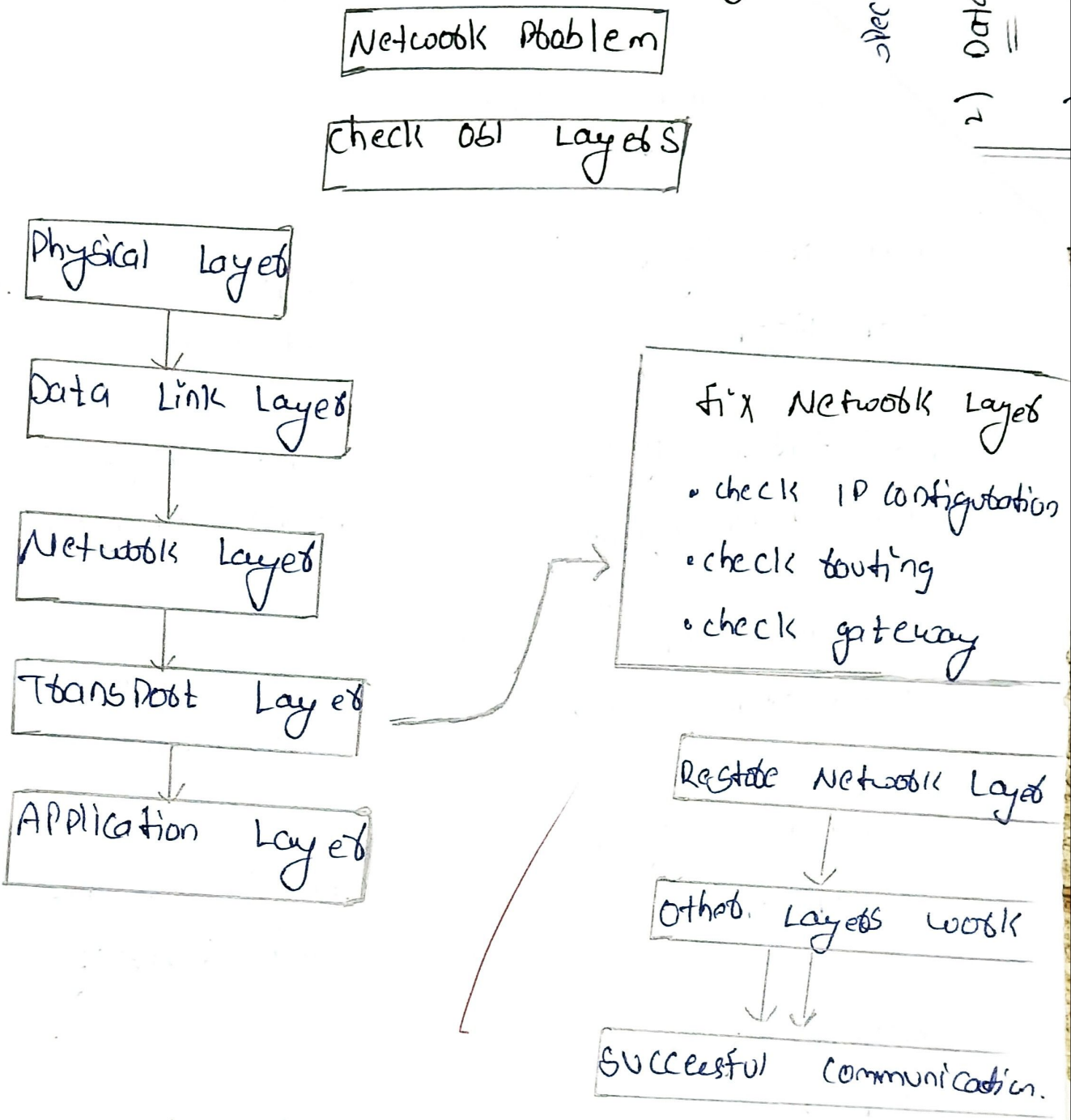
4) It handles data representation and format conversion

5) It helps ensure that data is represented in a suitable format.

Web Browsing using OSI Layers



3) Network Troubleshooting for



Explanation

1) The network cable or wi-fi connection is available

2) signals can travel between connected devices.

Therefore the Physical connection is the first suspected problem.

2) Data Link Layer - working

1) The local network connection is functioning

2) frames can be transferred over the local link.

3) Basic local communication is working.

3) Network Layer - failed

1) The Network Layer is responsible for IP address and routing

2) Therefore this layer should be investigated first

3) TCP connections may not be established successfully.